

Antifolate / nucleoside + PD-(L)1: 2 molecules stopped early, all small molecules

7 stopped trials, but 2 molecules — all small molecules: Pemetrexed and Gemcitabine. Against 4.8% for PD-(L)1 combinations without Antifolate / nucleoside and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

4.0%

7 of 176 closed trials (95% CI 1.9%–8.0%)

4 / 3 / 0

efficacy / safety / both (adds to 7)

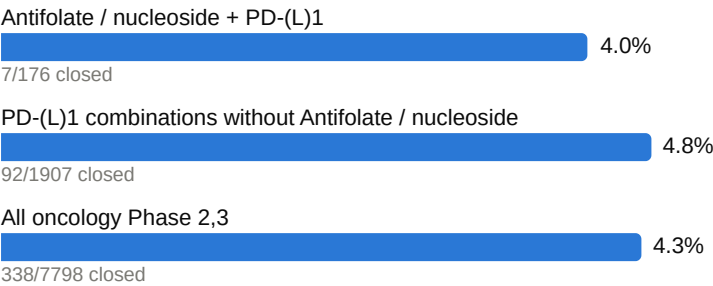
28.9%

of 610 trials have closed · 434 still open or unresolved

1.1%

stopped for a biological reason within 3 years of starting (95% CI 0.4%–3.0%; 235 still at risk; 1.5% area-wide)

How this compares



What is in this class: 310 distinct experimental drugs across 610 trials (small molecule 610, antibody 569, unknown 112, protein 51). Most tested: OXALIPLATIN (194), GEMCITABINE (189), CAPECITABINE (189), CISPLATIN (165), PACLITAXEL (148). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 7 stops came from 7 sponsor-asset programmes across 7 sponsors; the largest (Tufts Medical Center) contributed 1; removing that programme's trials from both sides leaves 3.4%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 19 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 14.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	44	2.3%	0.4%–11.8%
2018–2020	4	78	5.1%	2.0%–12.5%
2021–2024	2	54	3.7%	1.0%–12.5%

Where the stops sit

Efficacy / futility: 4, Safety: 3.
Sponsors with most stops: BerGenBio ASA (1), Tufts Medical Center (1), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (1), Nasser Hanna (1), Novartis (1), National Cancer Institute (NCI) (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05469178	2	BerGenBio ASA	BEMCENTINIB, PEMBROLIZUMAB, PEMETREXED, CARBOPLATIN	2025-04	Efficacy	Lack of efficacy
NCT04856176	2	Tufts Medical Center	Granulocyte-Macrophage Colony-Sti..., PEMBROLIZUMAB, PEMETREXED, PACLITAXEL	2024-12	Efficacy	Futility

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.
NCT03713944	2	Nasser Hanna	CARBOPLATIN, PEMETREXED, ATEZOLIZUMAB, BEVACIZUMAB	2021-12	Safety	Extreme toxicity, thromboembolic events
NCT03631199	3	Novartis 	CANAKINUMAB, PEMBROLIZUMAB, CARBOPLATIN, CISPLATIN	2021-08	Efficacy	Results of primary analysis showed addition of canakinumab to combination treatment did not improve tumor response or overall survival; the decision to stop the trial was not due to safety concerns
NCT03451773	2	National Cancer Institute (NCI)	BINTRAFUSP ALFA, GEMCITABINE	2020-05	Safety	Study was closed after one treatment related death.

The 6 most recent of 7 stopped trials. Every molecule behind all 7 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 28.9% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/data-licensing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.